

Fig. 2: Actual-size PCB pattern of the circuit for four frequencies generator

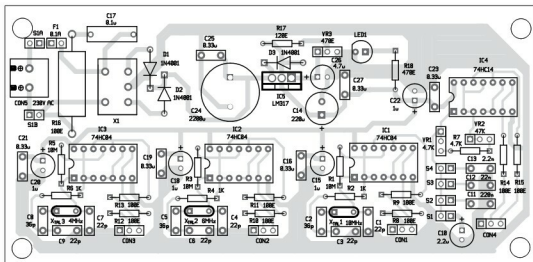


Fig. 3: Component layout of the PCB shown in Fig. 2

PARTS LIST

Semiconductors:

IC1-IC3	- 74HC04 hex inverting
IC4	- 74HC14 hex inverting schmitt trigger
IC5	- LM317, 3-terminal adjustable regulator
D1-D3	- 1N4001 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

R1, R3, R5	- 10-mega-ohm
R2, R4, R6	- 1-kilo-ohm
R7	- 4.7-kilo-ohm
R8-R15	- 100-ohm
R16	- 100-ohm, 2W
R17	- 120-ohm
R18	- 470-ohm
VR1	- 4.7-kilo-ohm potentiometer
VR2	- 47-kilo-ohm potentiometer
VR3	- 470-ohm potentiometer

Capacitors:

C1, C3, C4, C6, C7, C9	- 22pF ceramic disk
C2, C5, C8	- 36pF ceramic disk
C10	- 2.2µF, 16V electrolytic

C11	- 220nF ceramic disk
C12	- 22nF ceramic disk
C13	- 2.2nF ceramic disk
C14	- 220µF, 16V electrolytic
C15, C18, C20, C22	- 1µF, 16V electrolytic
C16, C19, C21, C23, C27, C25	- 0.33µF ceramic disk
C17	- 0.1µF, 630V polyester
C24	- 2200µF, 25V electrolytic
C26	- 4.7µF, 16V electrolytic

Miscellaneous:

CON1-CON4	- 3-pin connector terminal
CON5	- 2-pin connector terminal
X _{TAL1}	- 10MHz crystal
X _{TAL2}	- 6MHz crystal
X _{TAL3}	- 4MHz crystal
F1	- 2-pin connector terminal 0.1A fuse
S1-S4	- Toggle switch
S1A, S1B	- DPST switch
X1	- 230V AC primary to 15V-0-15V, 1A secondary transformer

IC3 uses crystal X_{TAL3} to produce the signal with a frequency of 4MHz. The produced output signal and inverted signal are available on connector CON3.

As a rule of thumb, R2 = reactance of C3 for IC1. Same goes for IC2 and IC3.

IC4 and associated components generate square wave with frequency from around 10Hz to more than 100kHz. Frequency ranges are selected with switches S1, S2, S3 and S4.

Potmeters VR1 and VR2 are used to adjust frequency. VR1 is for more accurate adjustment of frequency and VR2 for coarse adjustment of the frequency. The produced output signal and inverted signal are available on connector CON4. The frequency can be calculated according to the relationship given in the datasheet

of IC4.

Power supply for the oscillators should be regulated; this is done by adjustable regulator LM317. Trimmer potentiometer VR3 is used to adjust the power supply to 5V.

Construction and testing

An actual-size, single-side PCB for the four frequencies generator is shown in Fig. 2 and its component layout in Fig. 3. ●

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